

VPC-NAT Gateway

1. Create VPC in Europe Region

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

VPC-A

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
172.25.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

2. Create Subnet

Create subnet

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

pub-subnet
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

172.25.0.0/16

IPv4 subnet CIDR block
172.25.0.0/24 4,096 IPs

Tags - optional

Key	Value - optional
Name	pub-subnet

[Remove](#)

VPCSubnetsCreate subnet

Subnet 2 of 2

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

priv-subne

The name can be up to 256 characters long.

Availability Zone

Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block

Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

172.25.0.0/16

IPv4 subnet CIDR block

172.25.16.0/20

4,096 IPs

Tags - optional

Key

Value - optional

Q Name

Q priv-subne

Remove

Add new tag

You can add 49 more tags.

Remove

You have successfully created 2 subnets: subnet-043b0111381419f66, subnet-032dddea2f2d774a4

Subnets (1/2)

Info

Find subnets by attribute or tag

Subnet ID : subnet-043b0111381419f66

Subnet ID : subnet-032dddea2f2d774a4

Clear filters

	Name	Subnet ID	State	VPC
☐	priv-subne	subnet-032dddea2f2d774a4	Available	vpc-027b6da3630641104 V
☒	pub-subnet	subnet-043b0111381419f66	Available	vpc-027b6da3630641104 V

subnet-043b0111381419f66 / pub-subnet

Details

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

View details

Create flow log

Edit subnet settings

Edit IPv6 CIDRs

Edit network ACL association

Edit route table association

Edit CIDR reservations

Share subnet

Manage tags

Delete subnet

Subnet

Subnet ID

subnet-043b0111381419f66

Name

pub-subnet

Auto-assign IP settings

Info

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

Enable auto-assign public IPv4 address

Info

Enable auto-assign customer-owned IPv4 address

Info

Option disabled because no customer owned pools found.

Resource-based name (RBN) settings

Info

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

Enable resource name DNS A record on launch

Info

Enable resource name DNS AAAA record on launch

Info

Hostname type

Info

Resource name

IP name

3. Create & Attach Internet Gateway

An **Internet Gateway (IGW)** is a networking component in AWS that allows **your VPC** to **connect to the internet**.

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional** [Remove](#)

[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

Attach to VPC (igw-01025daca8f434ddb) [Info](#)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

► **AWS Command Line Interface command**

[Cancel](#) [Attach internet gateway](#)

4. Route table

A **Route Table** is a set of rules that decide **where your network traffic should go** inside a VPC.

Edit Routes

Route tables (1/2) [Info](#)

Last updated 11 minutes ago [Actions](#) [Create route table](#)

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Ow
☐	-	rtb-0a76891d23cfab0df	-	-	Yes	vpc-09b6bff8fa12c928	095
☒	VPC-A-route table	rtb-0e0ac9491c4a08813	-	-	Yes	vpc-027b6da3630641104 VPC-A	095

rtb-0e0ac9491c4a08813 / VPC-A-route table

[Details](#) [Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

Routes (1) [Both](#) [Edit routes](#)

Destination	Target	Status	Propagated	Route Origin
172.25.0.0/16	local	Active	No	Create Route Table

Launch an instance

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vpc

Create new key pair

Network settings

VPC - required

vpc-027b6da3630641104 (VPC-A)

Subnet

subnet-043b0111381419f66

pub-subnet

Create new subnet

Auto-assign public IP

Enable

Firewall (security groups)

Create security group

Select existing security group

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...read more

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Launch instance

Preview code

Private instance

Launch an instance

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vpc

Create new key pair

Network settings

VPC - required

vpc-027b6da3630641104 (VPC-A)

Subnet

subnet-032dddea2f2d774a4

priv-subnet

Create new subnet

Auto-assign public IP

Disable

Firewall (security groups)

Create security group

Select existing security group

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...read more

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Launch instance

Preview code

Instances (2)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
public-instance	i-0a205cc612c637b0e	Running	t3.micro	3/3 checks passed	View alarms	eu-north-1b	-
private-instance	i-09da4c2fa41282238	Running	t3.micro	Initializing	View alarms	eu-north-1b	-

While creating an instance, I generated a key pair, which is the private key. I need this private key to access my private-instance. Therefore, I will copy the private key to the public-instance using CMD, so I can connect from the public instance to the private instance.

```
Microsoft Windows [Version 10.0.26100.7171]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sjain403>cd downloads/

C:\Users\sjain403\Downloads>dir
Volume in drive C is OSDisk
Volume Serial Number is B29F-333D

Directory of C:\Users\sjain403\Downloads

26-Nov-25 03:29 PM <DIR> .
07-Nov-25 03:42 PM <DIR> ..
12-Nov-24 09:28 PM 3,073,800 (1).pdf
08-Apr-24 11:29 PM 228,754 (Prerana Badjate Resume).pdf
26-Jul-24 11:13 AM 60,304 0-1 Knapsack .pdf
06-Apr-24 02:13 PM 593,850 1027138186_WMHR25004961656_03042024.pdf
21-Jan-25 07:38 PM 680,591 1027138186_WMHR25008762668_02012025.pdf
06-Apr-25 12:05 PM 725,078 1027138186_WMHR26009924461_04042025.pdf
13-Oct-22 06:45 PM 1,281,502 10th Marksheet .pdf
13-Oct-22 06:51 PM 363,517 10th Marksheet _compressed.pdf
13-Mar-24 09:46 PM 291,040 11753806 (1).pdf
07-Jan-24 07:27 PM 260,458 11753806.pdf
17-Sep-25 08:03 AM 336,383 UTC APAC Server DAILY Backup Monitoring Logged 16th Sept 2025 1.xlsx
16-Sep-24 07:26 AM 824,649 UTC Exception 13th,14th,15th Sept 2024.xlsx
05-Jun-24 06:36 AM 6,146,044 UTC Monthly Report - May 2024.xlsx
17-Sep-25 08:01 AM 10,649,808 UTC- SUMMARY DBASE FOR 16th Sept 2026.xlsx
27-Jan-25 07:27 PM 9,547,087 VID-20231026-WA0007.mp4
16-Jan-24 07:26 PM 118,746 ViewResult1 (1).pdf
10-Jul-24 07:31 PM 119,512 ViewResult1 (11).pdf
10-Jul-24 08:01 PM 120,694 ViewResult1 (2).pdf
14-Jul-24 02:17 PM 119,174 ViewResult1 (3).pdf
01-Feb-25 12:27 PM 119,679 ViewResult1 (7) (1).pdf
24-Jan-25 07:42 PM 120,740 ViewResult1 (7).pdf
10-Jul-24 07:35 PM 120,689 ViewResult1.pdf
26-Nov-25 03:29 PM 1,674 vpc.pem
09-Feb-24 07:05 PM 98,950,632 VSCodeUserSetup-x64-1.86.1.exe
01-Apr-25 11:41 PM 34,655,352 WarpSetup.exe
14-Feb-25 08:06 PM 434,845 Web-Technologies.jpg
09-Aug-24 06:20 PM 33,843 WhatsApp Image 2024-08-09 at 6.20.36 PM.jpeg
15-Aug-24 04:07 PM 74,989 WhatsApp Image 2024-08-15 at 3.32.39 PM.jpeg
28-Mar-25 10:14 AM 981,048 WhatsApp Image 2025-03-28 at 10.14.40 AM.jpeg
28-Mar-25 10:17 AM 1,050,686 WhatsApp Image 2025-03-28 at 10.17.49 AM.jpeg
28-Mar-25 10:20 AM 992,606 WhatsApp Image 2025-03-28 at 10.20.04 AM.jpeg
29-Mar-25 05:34 PM 89,282 WhatsApp Image 2025-03-29 at 5.25.46 PM.jpeg
19-Apr-25 01:18 PM 1,413,753 WhatsApp Image 2025-04-16 at 8.32.25 PM.jpeg
```

A **Private Key** is used to securely login/authenticate to a remote system — especially without using a password.

```
C:\Users\sjain403\Downloads>scp -i vpc.pem vpc.pem ec2-user@13.60.32.157:/home/ec2-user
The authenticity of host '13.60.32.157 (13.60.32.157)' can't be established.
ED25519 key fingerprint is SHA256:97RkGZj7uYwY0mH1HrtT9gR9M4MPAUDNzW07TgHFgCU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added '13.60.32.157' (ED25519) to the list of known hosts.
vpc.pem 100% 1674 7.1KB/s 00:00

C:\Users\sjain403\Downloads>
```

Here I gave my public instance id

In Public-Instance you will see the key

```
aws
Q Search [Alt+S] Europe (Stockholm) Account ID: 0950-7410-7161 Sakshi Jain

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

(ec2-user@ip-172-25-9-122 ~)$ ls
vpc.pem
(ec2-user@ip-172-25-9-122 ~)$
```

Before taking access, for security I will give read only permission to root user

```
[ec2-user@ip-172-25-9-122 ~]$ chmod 400 vpc.pem
[ec2-user@ip-172-25-9-122 ~]$ ll vpc.pem
-r----- 1 ec2-user ec2-user 1674 Nov 26 10:29 vpc.pem
```

Now I will take access of Private Instance through SSH. Here I will use private instance Ip

```
[ec2-user@ip-172-25-9-122 ~]$ ssh -i vpc.pem ec2-user@172.25.21.194
```

```
#_
~\####_      Amazon Linux 2023
~~\_#####\
~~     \###|
~~       \#/   https://aws.amazon.com/linux/amazon-linux-2023
~~         V~' '->
~~~~
~~_.-./
~~/_m/'
```

```
[ec2-user@ip-172-25-21-194 ~]$
```

Now I will try to Install Package

```
[ec2-user@ip-192-168-18-250 ~]$ yum install mariadb105 -y
error: This command has to be run with superuser privileges (under the root user on most systems).
[ec2-user@ip-192-168-18-250 ~]$ sudo yum install mariadb105 -y
Amazon Linux 2023 repository | 100% | 0 B/s | 0 B --|-- ETA
```

This is my private instance there no network here so, I will use NAT Gateway this is providing network to private subnet.

6. NAT Gateway

A NAT Gateway (Network Address Translation Gateway) allows instances in a private subnet to access the internet but prevents the internet from accessing them.

Private EC2 instances (without public IPs) cannot reach:

- OS updates
- APIs
- Package downloads
- Software repositories

NAT Gateway solves this by:

- Allowing **outbound internet access only**
- Blocking **inbound internet traffic**

VPC > NAT gateways > Create NAT gateway

Elastic IP address 13.48.77.231 (eipalloc-0f487c01db700ed35) allocated.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

NAT gateway

The name can be up to 256 characters long.

Availability mode [Info](#)
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ Regional - new
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☒ Zonal
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.

subnet-043b0111381419f66 (pub-subnet)

Connectivity type
Select a connectivity type for the NAT gateway.

☒ Public

☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.

eipalloc-0f487c01db700ed35

[Allocate Elastic IP](#)

7. Create new Route table for Nat Gateway

AWS

VPC > Route tables > Create route table

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Route table-NAT

VPC
The VPC to use for this route table.

vpc-027b6da3630641104 (VPC-A)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q Route table-NAT

[Remove](#)

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create route table](#)

Edit Routes

AWS

VPC > Route tables > rtb-0879d951e852e3e76

rtb-0879d951e852e3e76 / Route table-NAT

[Actions](#)

Details [Info](#)

Route table ID
rtb-0879d951e852e3e76

Main
No

Explicit subnet associations
subnet-043b0111381419f66 / pub-subnet

Edge associations
-

VPC
vpc-027b6da3630641104 | VPC-A

Owner ID
095074107161

Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (1)

[Filter routes](#)

Destination	Target	Status	Propagated	Route Origin
172.25.0.0/16	local	Active	No	Create Route Table

[Both](#) [Edit routes](#)

< 1 >

[AWS](#)
[Search](#)
[\[Alt+S\]](#)
[Europe \(Stockholm\)](#)
[Account ID: 0950-7410-7161](#)
[Sakshi Jain](#)

[VPC](#) > [Route tables](#) > [rtb-0879d951e852e3e76](#) > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
172.25.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute

[Add route](#)
[Cancel](#)
[Preview](#)
[Save changes](#)

Select edit subnet associations

[AWS](#)
[Search](#)
[\[Alt+S\]](#)
[Europe \(Stockholm\)](#)
[Account ID: 0950-7410-7161](#)
[Sakshi Jain](#)

[VPC](#) > [Route tables](#) > [rtb-0e0ac9491c4a08813](#)

rtb-0e0ac9491c4a08813 / VPC-A route table

[Actions](#)

[Details](#)
[Info](#)

Route table ID
 rtb-0e0ac9491c4a08813

VPC
 vpc-027b6da3630641104 | VPC-A

Main
 Yes

Owner ID
 095074107161

Explicit subnet associations
 -

Edge associations
 -

[Routes](#)
[Subnet associations](#)
[Edge associations](#)
[Route propagation](#)
[Tags](#)

Explicit subnet associations (0)

[Find subnet association](#)
[Edit subnet associations](#)

No subnet associations
 You do not have any subnet associations.

Subnets without explicit associations (2)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

[Find subnet association](#)
[Edit subnet associations](#)

[VPC](#) > [Route tables](#) > [rtb-0879d951e852e3e76](#) > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

[Filter subnet associations](#)

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	priv-subnet	subnet-032dddea2f2d774a4	172.25.16.0/20	-	Main (rtb-0e0ac9491c4a08813 / VPC-...
☐	pub-subnet	subnet-043b0111381419f66	172.25.0.0/20	-	rtb-0e0ac9491c4a08813 / VPC-A-rout...

Selected subnets

subnet-032dddea2f2d774a4 / priv-subnet

[Cancel](#)
[Save associations](#)

8. Now you will get network to your private Ip

```

[ec2-user@ip-172-25-21-194 ~]$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=3.58 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=3.39 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=3.35 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=117 time=3.32 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=117 time=3.33 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=117 time=3.33 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=117 time=3.33 ms
    
```

```
[ec2-user@ip-172-25-21-194 ~]$ sudo yum install mariadb105 -y
Amazon Linux 2023 repository                                68 MB/s | 49 MB    00:00
Amazon Linux 2023 Kernel Livepatch repository             279 kB/s | 29 kB   00:00
Dependencies resolved.

Package Architecture Version Repository Size
-----
Installing:
mariadb105 x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 1.5 M
Installing dependencies:
mariadb-connector-c x86_64 3.3.10-1.amzn2023.0.1 amazonlinux 211 k
mariadb-connector-c-config noarch 3.3.10-1.amzn2023.0.1 amazonlinux 9.9 k
mariadb105-common x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 28 k
perl-Sys-Hostname x86_64 1.23-477.amzn2023.0.7 amazonlinux 16 k

Transaction Summary
-----
Install 5 Packages

Total download size: 1.8 M
Installed size: 19 M
Downloading Packages:
(1/5): mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch.rpm 281 kB/s | 9.9 kB 00:00
(2/5): mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64.rpm 4.3 MB/s | 211 kB 00:00
(3/5): mariadb105-10.5.29-1.amzn2023.0.1.x86_64.rpm 24 MB/s | 1.5 MB 00:00
(4/5): mariadb105-common-10.5.29-1.amzn2023.0.1.x86_64.rpm 1.0 MB/s | 28 kB 00:00
(5/5): perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64.rpm 903 kB/s | 16 kB 00:00
-----
Total 17 MB/s | 1.8 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded
```

Your database is installed successfully.